
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and local density removes most, half or little of it ($d = 20, 25,$
8 32). A second-order expansion of a linear map restricted to the manifold explains
9 why: a readout sees the second fundamental form contracted with its normal com-
10 ponent, $\langle w_N, \Pi \rangle$, an alignment’s second-order error is a connection mismatch plus
11 the relative second fundamental form, and mean curvature retains only the metric
12 trace of Π and determines neither. On 86,471 labelled galaxies a linear readout
13 gains accuracy where the manifold bends toward the label’s own curvature and
14 loses it where the manifold bends away. In a counterfactual local second-order
15 surrogate of the readout, the manifold’s bending in the readout’s normal direction
16 raises local R^2 at 79–99% of anchors, its sign reversal lowers it at 97–100%, and
17 a random normal direction with matched contracted norm does neither, on five
18 encoders; across anchors, the mismatch between the label Hessian and the probe-
19 facing Hessian is negative wherever the label has local structure and survives cross-
20 fitting and a change of encoder. Mean curvature of a single representation is there-
21 fore not a viable explanation for weak Platonic convergence; the obstruction is
22 relative and, for the fitted global linear alignment, already present at first order.

1 Introduction

24 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
25 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
26 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) share only 0.3 to
27 2.3 percent of nearest neighbours across surveys under a family of vision models. We ask where on
28 the manifold that agreement lives: if two models disagree most where the embedding surface bends
29 sharply, local mean curvature should associate negatively with local cross-survey agreement.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth; (ii) a density-controlled test of the curvature–
32 alignment prediction, which finds no effect free of density; (iii) the residual expansion of a linear
33 map on a submanifold, which names the curvature such a map responds to and shows mean-curvature
34 magnitude is blind to it; and (iv) known-surface, cross-encoder and interventional tests on the read-

out side: a linear readout gains local accuracy where the manifold bends toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor on five encoders.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. Embeddings are unit-normalized (data on S^{767}).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold has radial mean curvature $-d$, which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder. Decoder curvature has been a regularizer [Lee and Park, 2023] and an estimator [Acosta et al., 2023].

Validation. Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, correlated with $\|H_{\text{raw}}\|$ at $d \in \{20, 25, 32\}$.

Result. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009). But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density ($k = 30$ estimate on the Legacy embedding) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 (78%, 49% and 8% removed). Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; CKA [Kornblith et al., 2019] agrees), and removing the radial component of H_{raw} collapses it to -0.023 .

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(\|u\|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the rest, $\Pi = \Pi^S - g\hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x})g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaussian constant; real k -NN patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

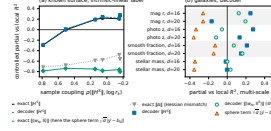


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder's $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05.

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order residual contains a connection mismatch and a *relative* extrinsic term, whose target-normal component is $\Pi_G - P_N^G A \Pi_F$.

So $\|H\|$, the norm of the metric trace of Π , need not correlate with either $\langle w_N, \Pi \rangle$ or the relative term; the second-order term scales as s^4 and, with fixed- k neighbourhoods, density sets s , so curvature effects scored on patches are density-sensitive.

5 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors' 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at five scales $k \in \{16, \dots, 2048\}$. Per anchor we compute $\|H^S\|$ and $\|\langle w_N, \Pi \rangle\|_g$, split by Eq. (2) into shape and sphere terms.

Known surface. Resampling the validation surface with weight $\|H^S\|^\gamma$ (Figure 1a) moves the $\|H^S\|$ partial from -0.29 to $+0.22$, reproduced by the decoder, while $\|\langle w_N, \Pi \rangle\|$ holds at -0.74 to -0.80 , there entirely the sphere term of Eq. (2).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\Gamma(x - x_0)$ of its 2,048 neighbours (a Monge chart: the quadratic coefficient is the covariant Hessian). Three regularities (Table 1; Appendix A). The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass (ridge shrinkage of extreme predictions). Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is null throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift): with $K = \langle w_N, \Pi \rangle$ it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so for infinitesimal scaling from zero the cross term decides whether the probe-facing curvature helps,

and at finite magnitude the $\|K\|^2$ term is the overshoot penalty. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55–0.65).

Bending toward helps, bending away hurts. By Eq. (3) curvature lowers the second-order residual relative to $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$: the manifold must bend toward the label’s own curvature, and not too far. (Antecedent: the optimal normal weight of least squares on a hypersurface, label second derivatives contracted with principal curvatures over their squares, derived as a uniqueness and stability condition [Liu et al., 2023]. Manifold capacity [Chung et al., 2016, Slatton et al., 2026] uses dimension and radius, flattening [Psenka et al., 2024] treats curvature as an obstacle, latent curvature tracks generalization [Kaufman and Azencot, 2023] and biases local linear regression [Cheng and Wu, 2013]; none conditions on the sign of $\langle \text{Hess}_M y, K \rangle$.) The partials compare anchors, and no anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing $K = K_S - (w \cdot \hat{x})g$ with $K_S = \langle w_S, \Pi^S \rangle$ the in-sphere shape term, each anchor’s 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2}K_{\text{sph}}(u, u) + \frac{t}{2}K_S(u, u)$ with a the readout’s tangent gradient, K_S from the decoder and c refit: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means $\langle R, K_S \rangle > 0$ for the remaining label curvature $R = \text{Hess}_M y - K_{\text{sph}}$), $t = -1$ its sign reversal, $t = 0$ shape-flat. The residual is exactly quadratic in t , so $t = 1$ beats shape-flat iff $2\langle R, K_S \rangle > \|K_S\|^2$ and $t^* = \langle R, K_S \rangle / \|K_S\|^2$. On five encoders and all four labels, $t = 1$ beats shape-flat at 79–99% of anchors, $t = -1$ is worse at 97–100%, and a random normal direction with $\|\langle v, \Pi^S \rangle\|_g$ matched to $\|K_S\|_g$ hurts at both signs ($t = 1$ beats shape-flat at 22–37% of anchors, $t^* \approx 0$; Appendix E). t^* sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (−0.24 to +0.27), as Eq. (1) allows. The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (−0.32 to −0.61; cross-fitted −0.28 to −0.62) and the alignment positive for magnitude on all five (+0.13 to +0.53) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (−0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed, at nearly every anchor and on five encoders. The residual expansion names the quantity, $\langle w_N, \Pi \rangle$ against $\text{Hess}_M y$, and a decoder’s pointwise second fundamental form measures it at production scale, validated against analytic truth for H^S and in substantial agreement with the data for the contracted Hessian. For the linear alignment map that recent accounts rest alignment on [Bangachev et al., 2026], the relevant second-order quantity is not one embedding’s mean curvature but $\Pi_G - P_N^G A \Pi_F$. Our pilot (Appendix C) finds a 44% first-order tangent-frame residual and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10–0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces (−0.14 to −0.88) but the alignment partial has no stable sign; Appendix E, not the fixture, ties alignment to local R^2 . Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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209 Appendix A: decoder ablations

210 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
 211 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
 variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	-0.26	-0.26	-0.27	-0.26
	mismatch	-0.39	-0.35	-0.34	-0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	-0.06*	-0.09*	-0.09	-0.06*
	sphere	-0.33	-0.35	-0.34	-0.35
	mismatch	-0.45	-0.43	-0.42	-0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	-0.13	-0.15	-0.14	-0.11
	mismatch	-0.09*	-0.05*	-0.08*	-0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	-0.06*	+0.05*	+0.09
	sphere	-0.35	-0.35	-0.35	-0.35
	mismatch	-0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	-0.02*	+0.03*	-0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

212

213 Appendix B: sensitivity of the mismatch and sphere columns

214 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local
 215 R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no
 216 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak
 217 ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which
 218 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that
 probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

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220 Appendix C: relative second fundamental form, pilot

221 Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A
 222 from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ;
 223 median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-
 224 controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of
 225 each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$, the same with L fitted

on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

Appendix E: intervening on the readout's normal component

The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-facing shape term hurts relative to removing it, since no anchor is flat. Here a counterfactual local second-order surrogate of the readout is scored at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; the anchor's 2,048 neighbours are scored by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the local intercept refit, where $q = \frac{1}{2} \langle w_S, \Pi^S \rangle (u, u)$ is the decoder's in-sphere second-order term in the anchor's chart coordinates: $t = 1$ is the manifold's bending as seen in the readout's normal direction, $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). The local sum of squares is an exact quadratic in t with minimum t^* ; with R the residual of the shape-flat readout, $t = 1$ beats shape-flat iff $2 \langle R, q \rangle > \langle q, q \rangle$. Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than shape-flat (hurt), the median change in local R^2

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

248 at $t = \pm 1$, the median t^* ; then the same for a random in-sphere normal direction v rescaled so that
249 $\|\langle v, \Pi^S \rangle\|_g = \|\langle w_S, \Pi^S \rangle\|_g$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction
250 of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; Supplement 12).

run	label	decoder in-sphere shape term					random normal direction, matched $\ \langle v, \Pi^S \rangle\ _g$			
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	−0.051	2.3	0.27	0.79	−0.009	−0.009
	photo_z	0.99	0.99	+0.028	−0.038	3.0	0.33	0.69	−0.005	−0.005
	smooth_fraction	0.90	1.00	+0.021	−0.040	1.7	0.22	0.81	−0.011	−0.012
	stellar_mass	0.96	0.99	+0.026	−0.040	2.6	0.26	0.73	−0.008	−0.007
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	−0.049	2.6	0.24	0.79	−0.007	−0.007
	photo_z	0.99	1.00	+0.029	−0.038	3.6	0.37	0.71	−0.003	−0.005
	smooth_fraction	0.90	1.00	+0.020	−0.035	1.9	0.22	0.78	−0.007	−0.009
	stellar_mass	0.97	1.00	+0.027	−0.038	3.1	0.32	0.71	−0.005	−0.005
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	−0.031	2.0	0.29	0.72	−0.007	−0.007
	photo_z	0.96	0.99	+0.020	−0.030	2.6	0.28	0.71	−0.006	−0.004
	smooth_fraction	0.93	1.00	+0.024	−0.055	1.5	0.22	0.77	−0.013	−0.012
	stellar_mass	0.95	1.00	+0.016	−0.028	2.1	0.27	0.73	−0.005	−0.005
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	−0.031	2.2	0.30	0.69	−0.006	−0.005
	photo_z	0.99	1.00	+0.019	−0.026	3.4	0.37	0.62	−0.002	−0.003
	smooth_fraction	0.92	0.99	+0.014	−0.027	1.7	0.22	0.72	−0.007	−0.006
	stellar_mass	0.97	1.00	+0.020	−0.028	2.8	0.35	0.68	−0.004	−0.005
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	−0.031	2.0	0.29	0.72	−0.006	−0.006
	photo_z	0.99	1.00	+0.018	−0.026	2.9	0.31	0.62	−0.004	−0.002
	smooth_fraction	0.88	0.98	+0.016	−0.036	1.6	0.22	0.77	−0.010	−0.008
	stellar_mass	0.99	1.00	+0.022	−0.032	3.1	0.29	0.63	−0.005	−0.003
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	−0.034	2.6	0.32	0.73	−0.005	−0.005
	photo_z	0.97	0.99	+0.018	−0.025	3.2	0.36	0.64	−0.003	−0.003
	smooth_fraction	0.79	0.97	+0.013	−0.029	1.6	0.25	0.77	−0.007	−0.008
	stellar_mass	0.98	1.00	+0.022	−0.032	2.9	0.32	0.66	−0.003	−0.003

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at nearly every anchor, its sign reversal hurts at nearly every anchor, a random normal direction with matched contracted norm does neither; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.